Class 5: Data Viz with ggplot

Barry (PID: 911)

Today we are exploring the **gggplot** package and how to make nice figures in R.

There are lots of ways to make figures and plot in R. These include:

- so called "base" R
- and add on packages like **ggplot2**

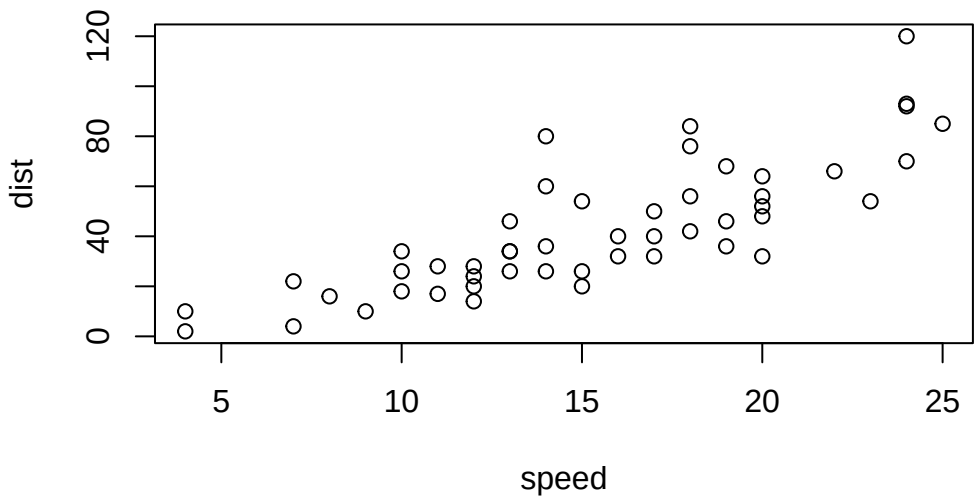
Here is a simple "base" R plot.

```
head(cars)
```

```
  speed dist
1     4    2
2     4   10
3     7    4
4     7   22
5     8   16
6     9   10
```

We can simply pass to to the `plot()` function.

```
plot(cars)
```



Key-point: Base R is quick but not so nice looking in some folks eyes.

Let's see how we can plot this with **ggplot2**…

1st I need to install this add-on package. For this we use the `install.packages()` function - **WE DO THIS IN THE CONSOLE, NOT our report**. This is a one time only deal.

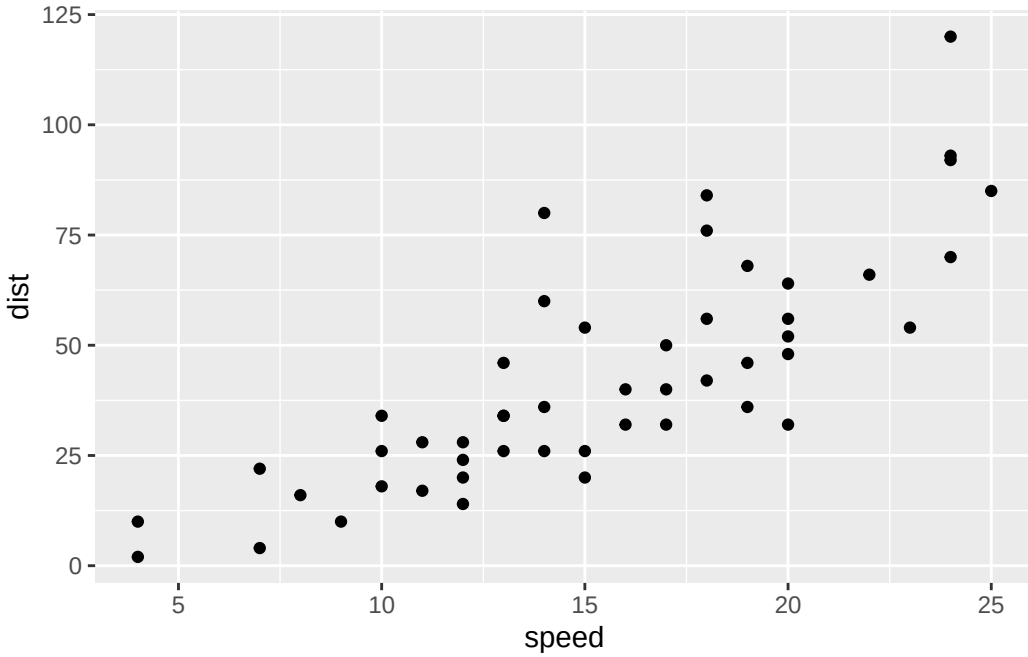
2nd We need to load the package with the `library()` function every time we want to use it.

```
library(ggplot2)
ggplot(cars)
```

Every ggplot is composed of at least 3 layers:

- **data** (i.e a data.frame with the things you want to plot),
- aesthitics **aes()** that map the columns of data to your plot features (i.e. aesthitics)
- geoms like **geom_point()** that srt how the plot appears

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point()
```



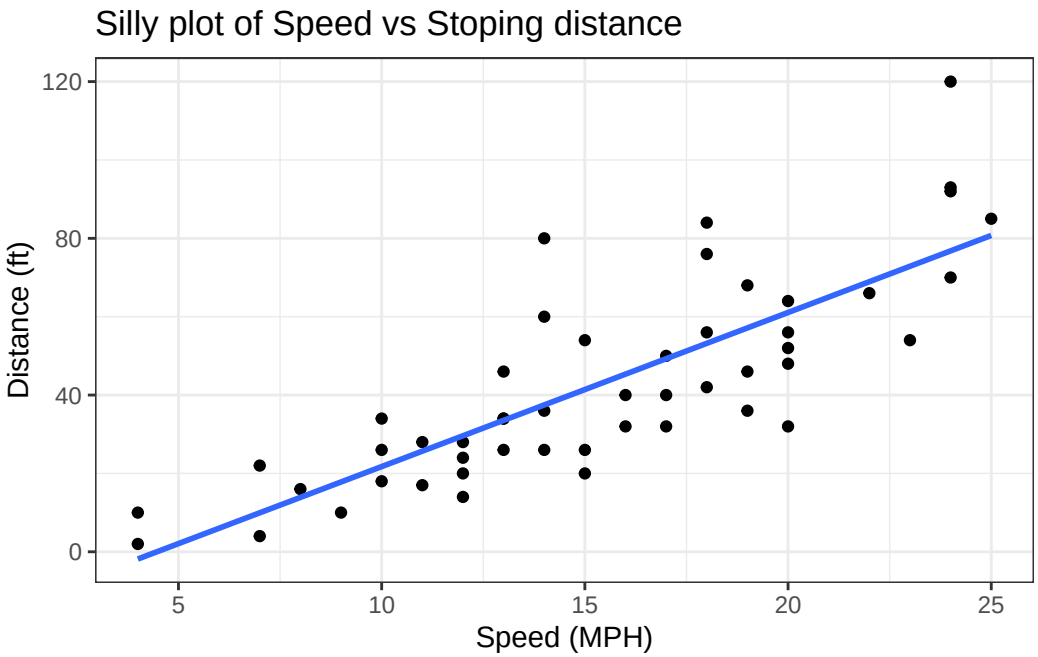
> Key point: For simple "canned" graphs base R is quicker but as things get more custom and elobrate then ggplot wins out…

Let's add more layers to our ggplot

Add a line showing the relationship between x and y Add a title Add custom axis labels "Speed (MPH)" and "Distance (ft)" Change the theme…

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE) +
  labs(title="Silly plot of Speed vs Stoping distance",
       x="Speed (MPH)",
       y="Distance (ft)") +
  theme_bw()
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Going further

Read some gene expresion data

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)

head(genes)
```

```
        Gene Condition1 Condition2      State
1      A4GNT -3.6808610 -3.4401355 unchanging
2       AAAS  4.5479580  4.3864126 unchanging
3      AASDH  3.7190695  3.4787276 unchanging
4       AATF  5.0784720  5.0151916 unchanging
5       AATK  0.4711421  0.5598642 unchanging
6 AB015752.4 -3.6808610 -3.5921390 unchanging
```

Q1. How many genes are in this wee dataset?

```
nrow(genes)
```

```
[1] 5196
```

```r
ncol(genes)
```

```
[1] 4
```

Q2. How many "up" regulated genes are there?

```r
sum( genes$State == "up" )
```

```
[1] 127
```

A useful function for counting up occurances of things in a vector is the `table()` function.

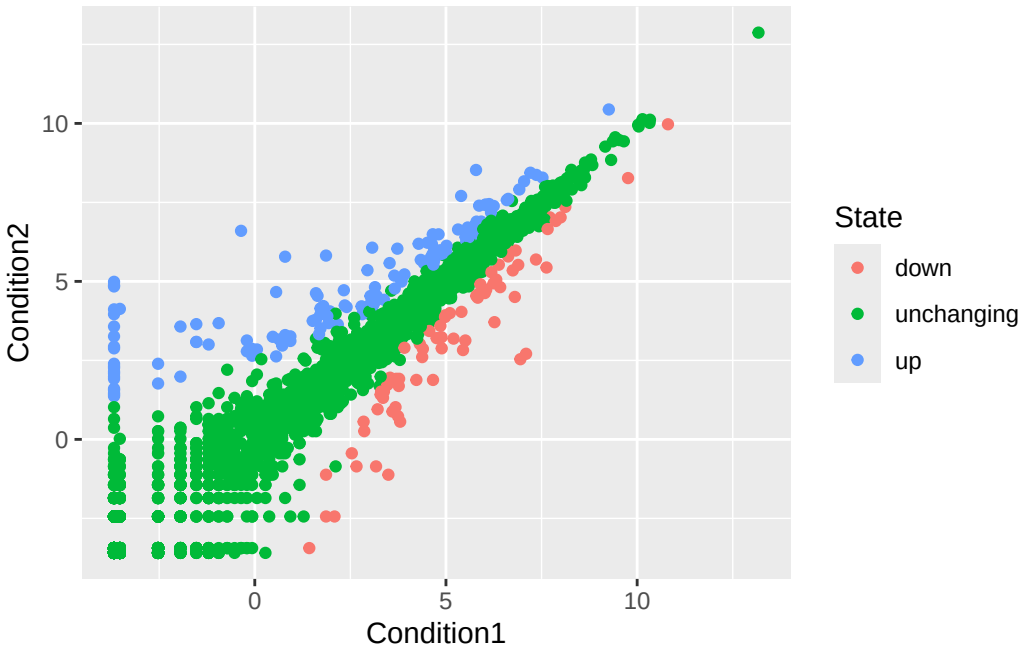
```r
table( genes$State )
```

```
      down unchanging         up 
        72       4997        127 
```

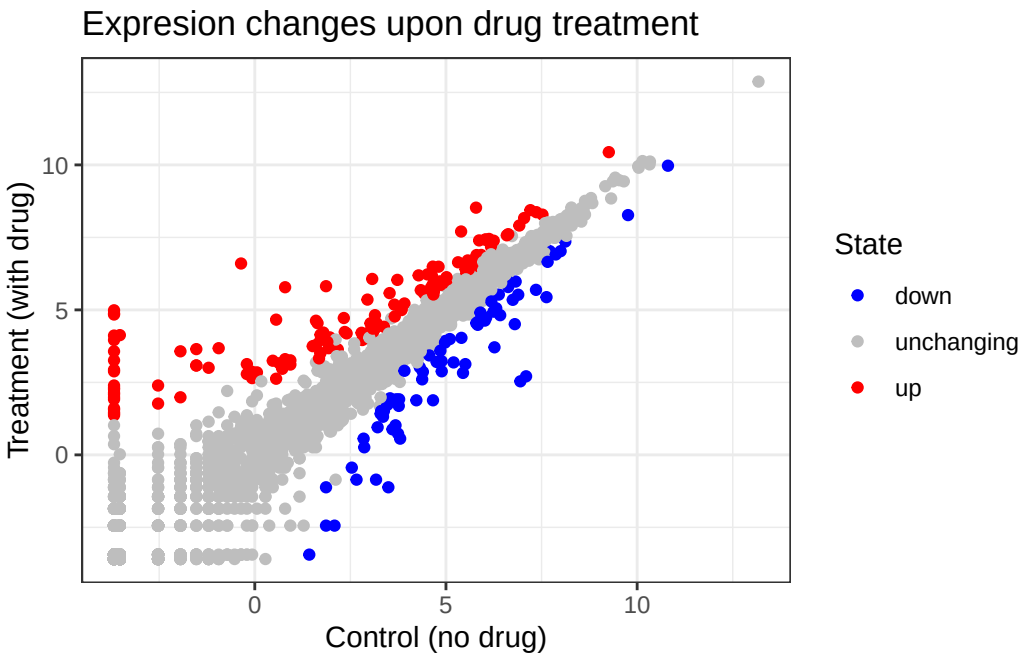
Make a v1 figure

```r
p <- ggplot(genes) +
  aes(x=Condition1,
      y=Condition2,
      col=State) +
  geom_point()

p
```




```
p +
  scale_colour_manual( values=c("blue","gray","red") ) +
  labs(title="Expresion changes upon drug treatment",
       x="Control (no drug)",
       y="Treatment (with drug)") +
  theme_bw()
```



More Plotting

Read in the gapminder dataset

```
# File location online
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv

gapminder <- read.delim(url)
```

Lets have a wee peak

```
head( gapminder, 3)
```

```
      country continent year lifeExp      pop gdpPercap
1 Afghanistan      Asia 1952  28.801  8425333  779.4453
2 Afghanistan      Asia 1957  30.332  9240934  820.8530
3 Afghanistan      Asia 1962  31.997 10267083  853.1007
```

```
tail(gapminder, 3)
```

```
      country continent year lifeExp      pop gdpPercap
1702 Zimbabwe    Africa 1997  46.809 11404948  792.4500
1703 Zimbabwe    Africa 2002  39.989 11926563  672.0386
1704 Zimbabwe    Africa 2007  43.487 12311143  469.7093
```

Q4. How many different country values are in this dataset?

```r
nrow(gapminder)
```

```
[1] 1704
```

```r
length( table(gapminder$country) )
```

```
[1] 142
```

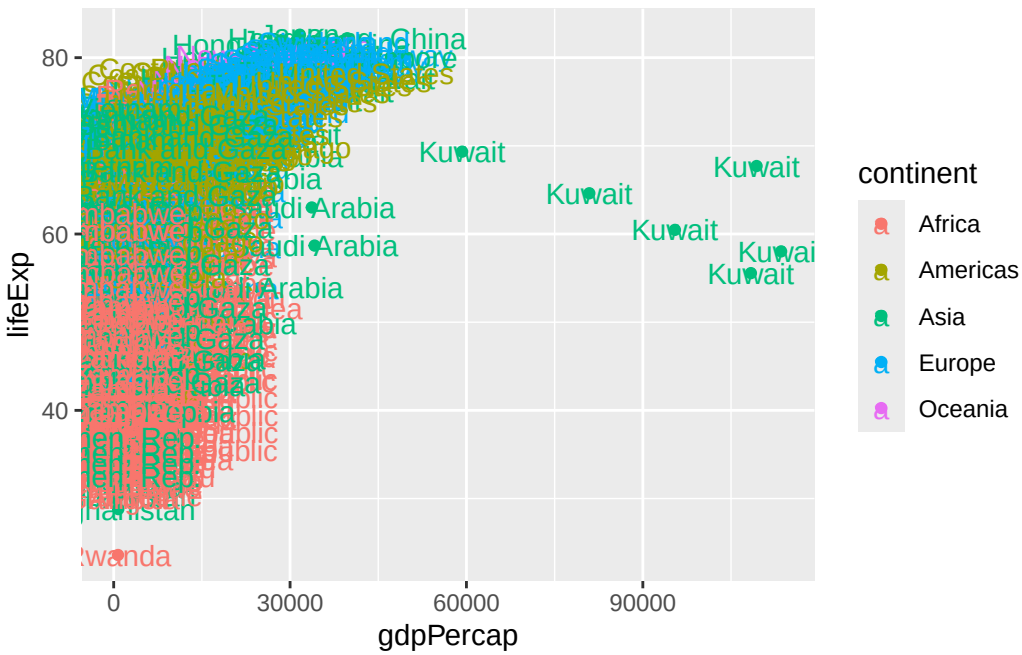
Q5. How many different continent values are in this dataset.

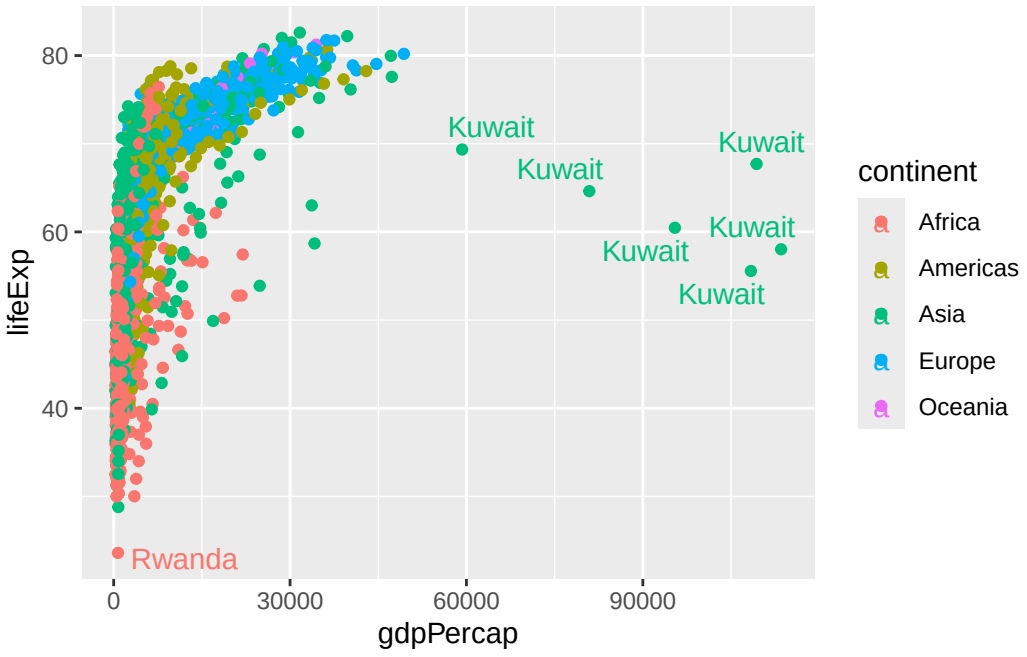
```r
unique(gapminder$continent)
```

```
[1] "Asia"     "Europe"   "Africa"   "Americas" "Oceania"
```

```r
ggplot(gapminder) +
  aes(gdpPercap, lifeExp, col=continent) +
  geom_point()
```

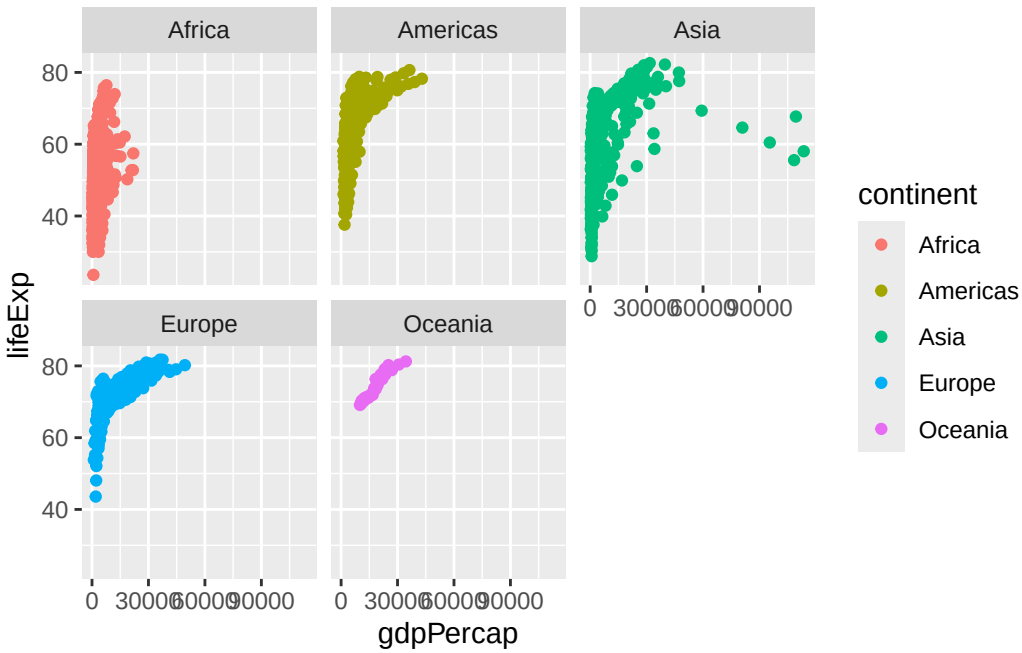
```r
ggplot(gapminder) +
  aes(gdpPercap, lifeExp, col=continent, label=country) +
  geom_point() +
  geom_text()
```


lifeExp
80
60
40
0
30000
60000
90000
gdpPercap
Kuwait
Kuwait
Kuwait
Kuwait
Kuwait
Kuwait
Kuwait
Kuwait
Arabia
Arabia
Arabia



I want a seperate pannel per continent

```
ggplot(gapminder) +
  aes(gdpPercap, lifeExp, col=continent, label=country) +
  geom_point() +
  facet_wrap(~continent)
```



Summary

The main advantages of ggplot over base R plot are:

1. **Layered Grammar of Graphics**: ggplot uses a consistent, layered approach where you build plots by adding layers for data, aesthetics, and geometries. This makes complex plots easier to construct and modify, compared to base R which often requires different functions and many arguments for each plot type [1], [2], [3], [5], [4].
2. **Publication-Quality Output**: ggplot produces visually appealing, publication-ready figures by default, while base R plots often need extensive tweaking to look polished [1], [2], [3], [4].
3. **Declarative Syntax**: You specify what you want to show (data, mappings, geoms) rather than how to draw each element, making code more readable and maintainable [1], [2], [3], [5], [4].
4. **Easy Customization and Extension**: Adding titles, axis labels, legends, colors, and themes is straightforward and consistent. In base R, these customizations can be fiddly and require more manual control [1], [2], [3], [5], [4].
5. **Reproducibility**: ggplot code is easy to reuse and adapt for new datasets or visualizations, supporting reproducible research [1], [2], [3], [4].

Which of these advantages do you think is most important for your work? Why?